

COMPUTER VISION

ASSIGNMENT 1

07 AUGUST 2026

Q1. Image Resizing, Interpolation, and Edge Analysis

Select a standard grayscale image of size 512×512 pixels and perform the following operations:

1. Downsample the original image to 256×256 and 128×128 pixels.
2. Resize each downsampled image back to 512×512 pixels using:
 - Nearest-neighbour interpolation
 - Bilinear interpolation
 - Bicubic interpolation
3. Compare the reconstructed images with the original image using MSE and PSNR.
4. Display the pixel-wise absolute-error and squared-error maps for each reconstructed image.
5. Compare the interpolation methods in terms of visual quality, reconstruction error, and edge preservation.
6. Tabulate all results and identify the best-performing interpolation method.

Note: The effect of the downsampling factor on image reconstruction and edge preservation must be discussed. At least one interpolation method should be implemented without using a built-in image-resizing function.

Q2. Image Smoothing

Select two grayscale images: one containing fine details and another containing large smooth regions. Add salt-and-pepper noise to the first image and Gaussian noise to the second image. Perform the following tasks:

1. Smooth both noisy images using Box, Weighted-average, Gaussian, and Median filters.

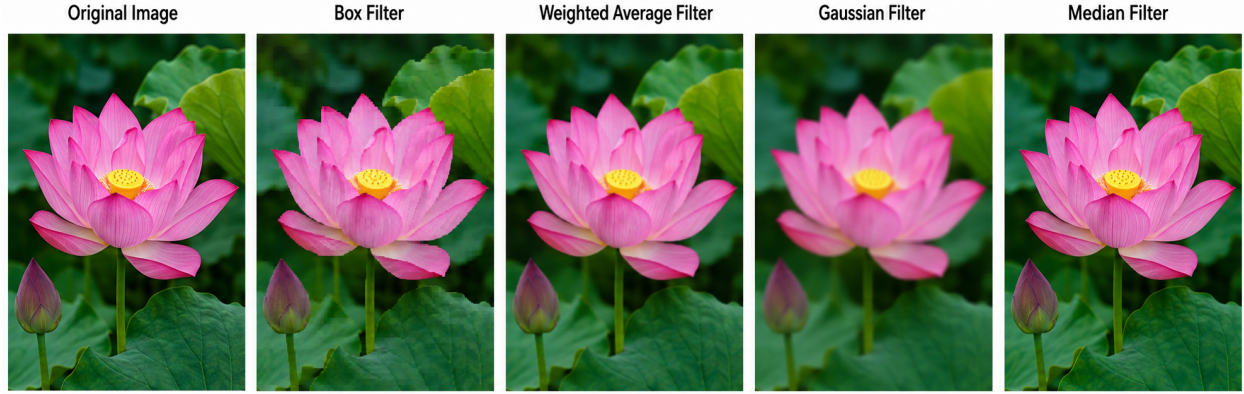


Figure 1: Comparison of image-smoothing methods.

2. Apply each filter using at least three different kernel sizes and observe the effect of increasing the kernel size.
3. Compare the filtered images in terms of noise removal, edge preservation, detail loss, and blurring.
4. Calculate MSE and PSNR between each filtered image and its original noise-free image.
5. Display an absolute-difference image showing the information removed by each filter:

$$D(x, y) = |I(x, y) - I_f(x, y)|,$$

where I and I_f represent the noisy and filtered images, respectively.

6. Create a single corrupted image whose left half contains salt-and-pepper noise and right half contains Gaussian noise. Apply all four filters and identify whether one filter performs best over the complete image.
7. Design a simple region-wise smoothing approach that applies the best-performing filter to each half of the mixed-noise image.
8. Tabulate the results and recommend the most suitable filter for each noise type.

Note: At least one smoothing filters must be implemented without directly using built-in filtering functions. The selected parameters and conclusions must be justified.

Q3. Hybrid Images

A hybrid image combines the low-spatial-frequency information of one image with the high-spatial-frequency information of another. Its interpretation changes with viewing distance: the high-frequency image is generally dominant when viewed closely, whereas the low-frequency image becomes more prominent when viewed from a distance or after the image is downsampled.

For two aligned images I_1 and I_2 , a hybrid image can be generated as

$$H = \alpha \text{LPF}(I_1) + \beta \text{HPF}(I_2), \quad (1)$$

where

$$\text{HPF}(I_2) = I_2 - \text{LPF}(I_2). \quad (2)$$

while α and β control their relative contributions. Perform the following tasks:



Figure 2: An example of hybrid images that combine the low-spatial-frequency information of one facial expression with the high-spatial-frequency information of another. When viewed up close, the faces appear sad; however, when viewed from a few metres away, their expressions appear to change.

1. Select at least two image pairs and align their important features using suitable geometric transformations.
2. Generate hybrid images using Gaussian low-pass and high-pass filters in the spatial domain.
3. Experiment with different filter sizes, standard deviations, and component weights. Select the parameters that produce the best hybrid image.
4. Implement the same filtering in the frequency domain using the two-dimensional Fourier transform.
5. Compare the spatial- and frequency-domain results in terms of visual quality and execution time.
6. Study and make use of Gaussain Pyramids and Laplacian Pyramids for mixing the two images.
7. Implement Laplacian Pyramids with Bilateral filters and use them with Gaussain Pyramids for the same.
8. Briefly discuss how image alignment and frequency separation influence the quality of a hybrid image.

Note: High-pass-filtered images contain both positive and negative values. Normalize them only for visualization. Use the original filtered values when constructing the hybrid image.

Q4. Edge Detection and Analysis

Select two grayscale images: one containing clearly defined object boundaries and another containing fine details. Perform the following operations:

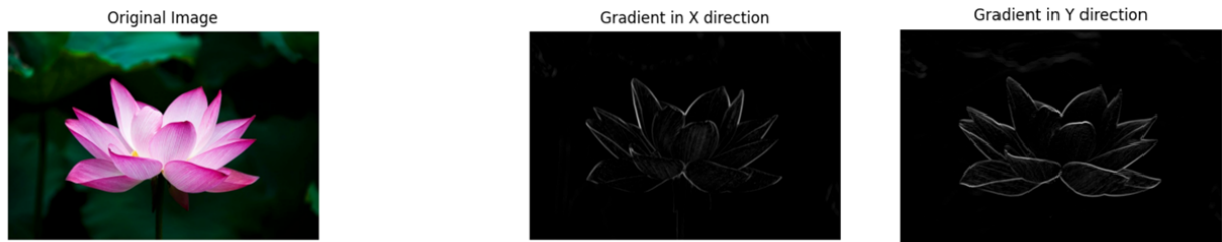


Figure 3: Illustration of first-order image derivatives. From left to right: the original image, gradient response in the x -direction, and gradient response in the y -direction.

1. Compute the first-order derivatives along the x - and y -directions.
2. Calculate and display the gradient magnitude and gradient direction.
3. Generate binary edge maps using three different threshold values. Discuss how the threshold affects missing and unwanted edges.
4. Generate an edge map using the second-order derivative and compare it with the first-order derivative result.
5. Generate edge maps using the Laplacian of Gaussian (LoG) and Canny edge detectors.
6. Compare the obtained edge maps in terms of edge localization, continuity, thickness, and detection of fine details.
7. Identify the most suitable edge-detection method for each selected image and justify your answer.

Note: Display all intermediate and final outputs. The first- and second-order derivative operations must be implemented without directly using built-in edge-detection functions.

Evaluation: Original implementations and creative image combinations will be awarded higher marks. Justification of the selected parameters will also be evaluated.